

AXIOM A1

PT-2 GROUNDED

SOURCE: [PT2-Q10, Q14]

EVIDENCE: [SW-Q10, Q14]

{{ LESSON_TITLE: Equality is a Promise - Legal Coordinate & Algebraic Transformations }}

Cognitive Job: Master the invariant preservation principle before memorizing formulas. The '=' sign is a promise.

{{ AXIOMATIC_INVARIANT: Equality is an Unbroken Contract }}

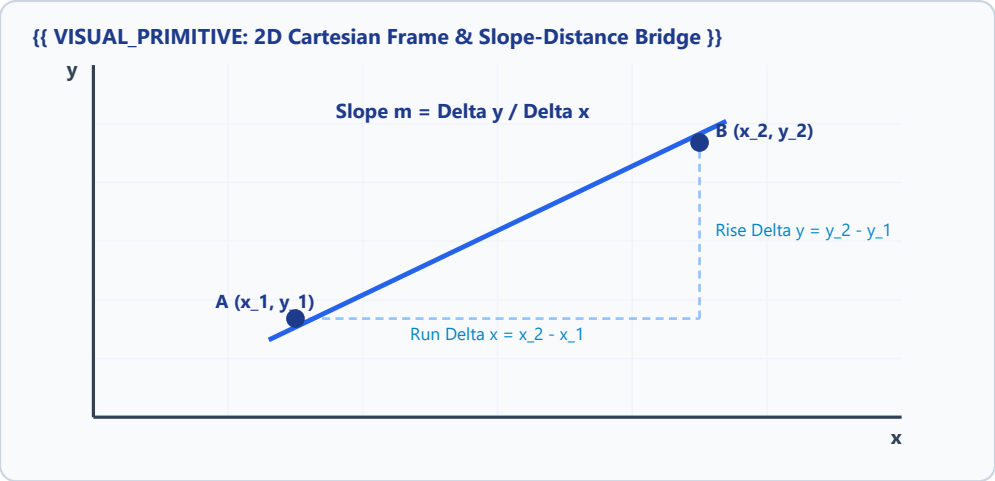
The equals sign does not signify 'now compute the next line'.

It asserts that the expression on the left and the expression on the right evaluate to the exact same mathematical value or geometric point.

- * Legal Rule 1: Applying any bijection $f(x)$ to both sides preserves equality: $A = B \Rightarrow f(A) = f(B)$.
- * Legal Rule 2: Compound terms (such as $x + y$) must be bracketed before transformation.
- * Invariant Check: If a candidate step produces unequal evaluations, the step is ILLEGAL.

{{ MISCONCEPTION_CLINIC: Why the Learner's Q14 Route Failed }}

- * The Error: Student computed $\text{sqrt}(x_1^2 + y_1^2)$ without grouping differences, missing the true distance invariant: $d = \text{sqrt}((x_2 - x_1)^2 + (y_2 - y_1)^2)$.
- * Root Cause: Misapplied square root across addition: $\text{sqrt}(a + b) \neq \text{sqrt}a + \text{sqrt}b$.
- * Causal Repair: Distance is the hypotenuse of a right-angled slope triangle. You must calculate the horizontal run Delta x and vertical rise Delta y first, then apply Pythagoras to the components: $d^2 = (\text{Delta } x)^2 + (\text{Delta } y)^2$.



WORKED EXAMPLE

{{ WORKED_EXAMPLE_TITLE: Line Passing Through (4, -1) with Slope 2/3 }}

Problem: Find the equation of the line passing through A(4, -1) with slope $m = 2/3$, and determine its y-intercept.

Reasoning Story (Step-by-Step Invariant Preservation):

1. State the Definition: Slope is constant for every point (x, y) on the line:
 $(y - y_1) / (x - x_1) = m \Rightarrow (y - (-1)) / (x - 4) = 2/3$
2. Preserve Compound Equality (Multiply both sides by $3(x - 4)$):
 $3(y + 1) = 2(x - 4)$
 $3y + 3 = 2x - 8$
3. Canonical Standard Form: $2x - 3y - 11 = 0$
4. Invariant Check (Substitute $x = 4, y = -1$): $2(4) - 3(-1) - 11 = 8 + 3 - 11 = 0$. [VALIDATED]
5. Intercept: Set $x = 0 \Rightarrow 3y = -11 \Rightarrow y = -11/3$.

TRY WITH ME * FADED PRACTICE

{{ FADED_PRACTICE_Q1 }}: The line passes through P(-2, 3) and Q(4, -1).

Step 1: Compute Rise and Run: Delta y = $(-1) - 3 = \underline{\hspace{1cm}}$ | Delta x = $4 - (-2) = \underline{\hspace{1cm}}$

Step 2: Slope $m = \text{Delta } y / \text{Delta } x = \underline{\hspace{1cm}} / \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

Step 3: Point-Slope Form: $y - (\underline{\hspace{1cm}}) = m(x - (\underline{\hspace{1cm}}))$

Step 4: Check your answer by testing whether point Q(4, -1) satisfies your equation.

{{ FADED_PRACTICE_Q2 }}: State Euclid's Common Notion 1 in terms of legal algebra:

'Things which are equal to the same thing are equal to one another.'

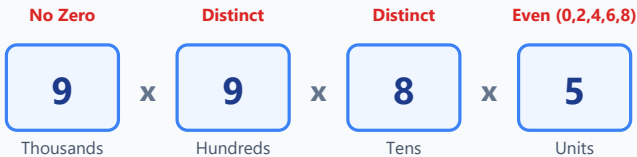
If $A = C$ and $B = C$, prove why $A = B$ using a single substitution step.

Your deduction: _____

DISCRETE MATH
 COUNTING ENGINE
 ST01-ST12 COVERAGE

{{ LESSON_TITLE: The Multiplication Principle as Independent Decision Slots }}

{{ VISUAL_PRIMITIVE: 4-Digit Number Formation Slot Model }}



{{ COMBINATORIAL_RULES: Slot Restrictions & Priority Order }}

1. The Severest Restriction Goes First:

If a slot has special conditions (e.g. 'Units digit must be even', 'Thousands != 0'), fill that decisive slot FIRST before unconstrained positions.

2. Distinguish With vs Without Repetition:

- * With Repetition: Previous choices do not reduce subsequent options: $n \times n \times n$.
- * Without Repetition: Each filled slot consumes 1 object: $n \times (n - 1) \times (n - 2)$.

3. Independent Decisions Rule (Rule of Product):

If task 1 can be done in p ways, and for each way task 2 can be done in q ways, the combined sequence occurs in $p \times q$ ways.

4. Fail-Safe Overcounting Check: Check whether indistinguishable objects were ordered as if distinguishable. Divide by $k!$ to remove unwanted symmetry.

COMBINATORICS WORKED EXAMPLE

{{ EXAMPLE: 4-Digit Even Numbers Without Repetition }}

Problem: How many 4-digit even numbers can be formed using digits {0, 1, 2, 3, 4, 5} if no digit may be repeated?

Reasoning Story (Split by Conflicting Restriction):

The digit '0' creates a conflict: it makes a number even (units slot), but cannot be in the thousands slot. Therefore, split into 2 mutually exclusive cases:

Case 1: Units digit is 0 (Units = {0} => 1 way):

- * Thousands: Any non-zero remaining digit {1,2,3,4,5} => 5 choices.
- * Hundreds: Any remaining digit => 4 choices.
- * Tens: Any remaining digit => 3 choices.
- * Subtotal Case 1 = $5 \times 4 \times 3 \times 1 = 60$ numbers.

Case 2: Units digit is non-zero even (Units = {2, 4} => 2 ways):

- * Thousands: Cannot be 0, cannot be the units digit => $(6 - 2) = 4$ choices.
- * Hundreds: Remaining digits (including 0 now allowed) => 4 choices.
- * Tens: Remaining digits => 3 choices.
- * Subtotal Case 2 = $4 \times 4 \times 3 \times 2 = 96$ numbers.

Total = Case 1 + Case 2 = 60 + 96 = 156 even numbers. [CHECKED & VERIFIED]

CORE 1 -> CORE 2 TRANSFER CONTRACT

{{ CORE2_TRANSFER_AUTHORITY }}

- * Core 1 establishes the mental models, invariants, and failure clinics.
 - * Core 2 puts you in full test conditions with untouched external items.
 - * When stuck in Core 2, use the H1-H3 Hint Ladder before checking solutions.
 - * Every Core 2 question explicitly cites its foundational Core 1 lesson.
- Linked Target: Core 2 Set ST01 - Elementary Slot Permutations (Questions 1-15)